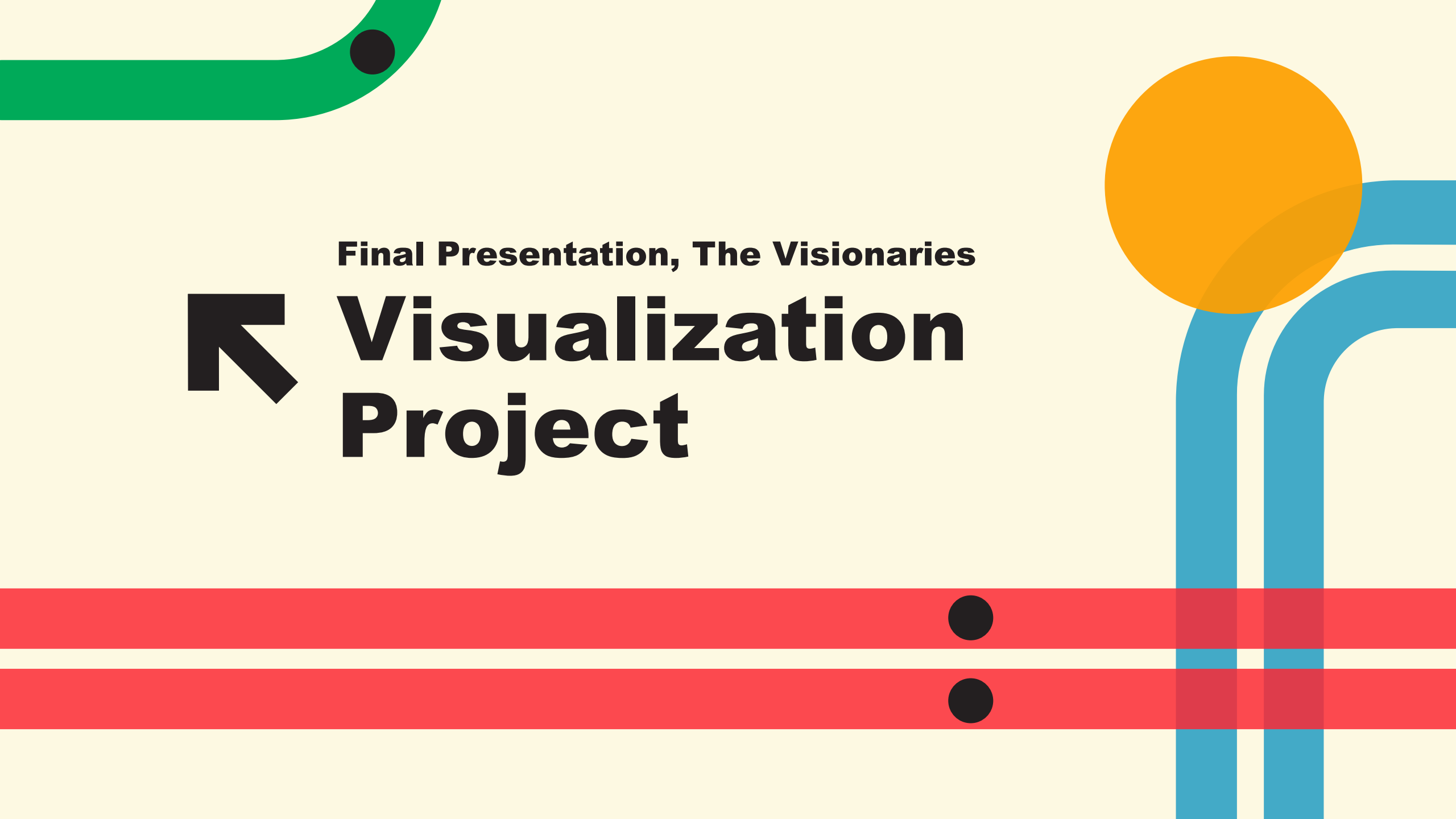
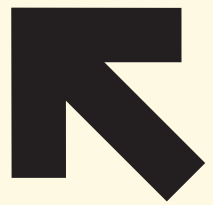




Final Presentation, The Visionaries



Visualization Project



2

Introduction

Team Members:

- **Aaron Leeb – Team Lead**
- **Koleman Lund – Tech Lead**
- **Trevor Cloutier – Sankey Specialist**
- **Michael Asmah – Data Designer**

Faculty:

- **Dr. Jonathan Hardwick – Faculty Coach**
- **Ryne Leytem – Subject Matter Expert**
- **Dr. Emily Gresbrink – Tech-Com Coach**
- **Other CS Faculty – SMEs/Resources**

Agenda

Project Overview

1

Demo

2

Technical Overview

3

Work Completed

4

Deliverables & Timeline

5

Successes / Challenges

6

The background features a light cream color with several abstract geometric elements. A thick red line enters from the top right, turns 90 degrees left, and extends horizontally. A thick green line enters from the left, curves 90 degrees down, and then turns 90 degrees right, continuing horizontally. A thick blue line enters from the bottom, curves 90 degrees up, and then turns 90 degrees left, continuing horizontally. An orange circle is positioned on the left side, between the green and blue lines. Two small black dots are located on the green line: one at its initial downward curve and another at its intersection with the blue line.

Project Overview



5

Our Client

Dr. Lin Chase

- ***Executive Director of Institutional Research and Planning***
- **Minnesota State University, Mankato**

Dr. Shauna Elbers

- ***Associate Provost for Faculty Affairs & Learning Innovation***
- **Minnesota State University, Mankato**



6

Goal

**Large amount of
MNSU student
historical data**



**Create a user-
friendly
application that
displays this data**



7

Purpose

Build a tool to help faculty see trends of student progress throughout their time.

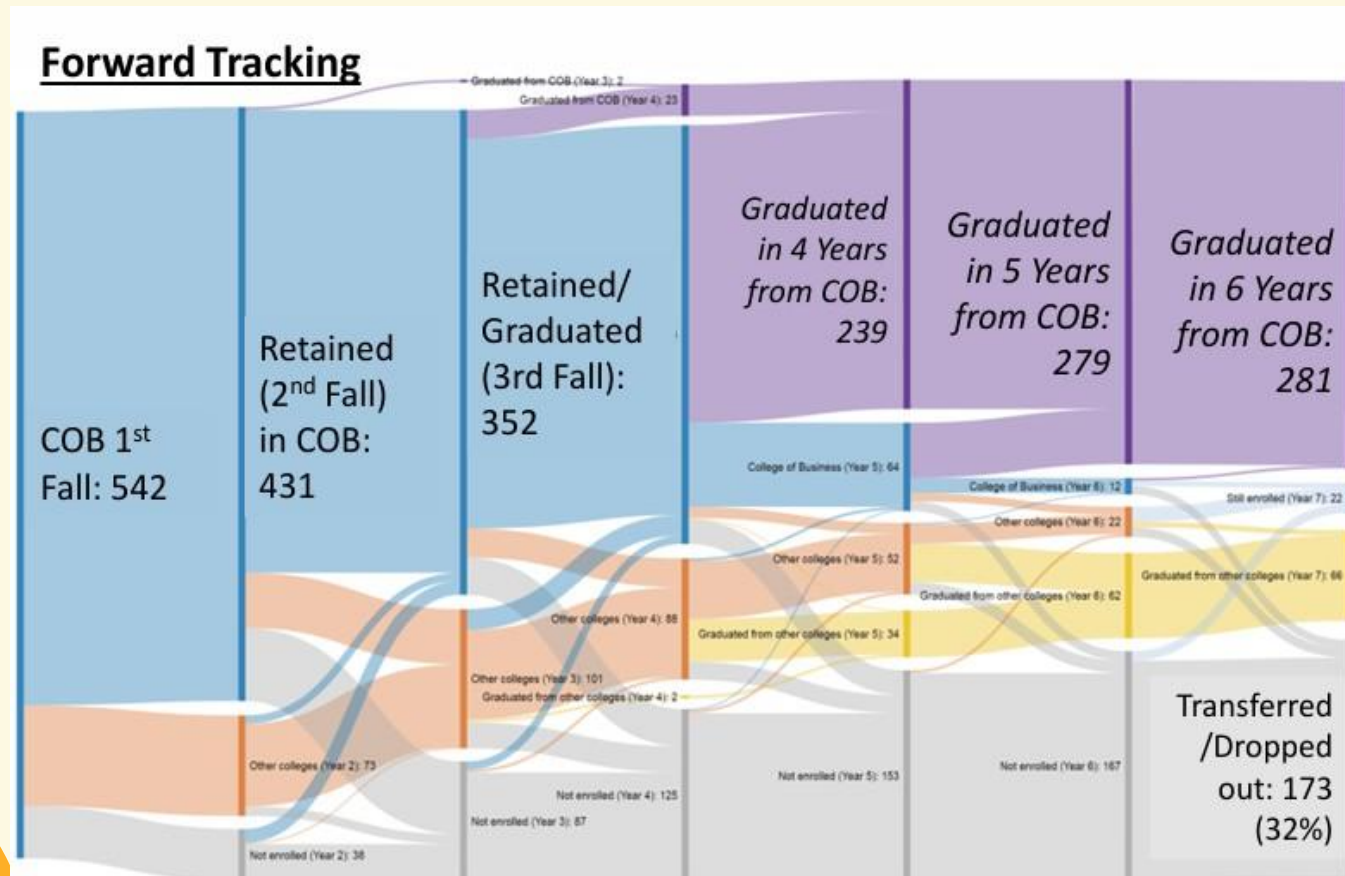


<https://mankato.mnsu.edu/giving/foundation/>

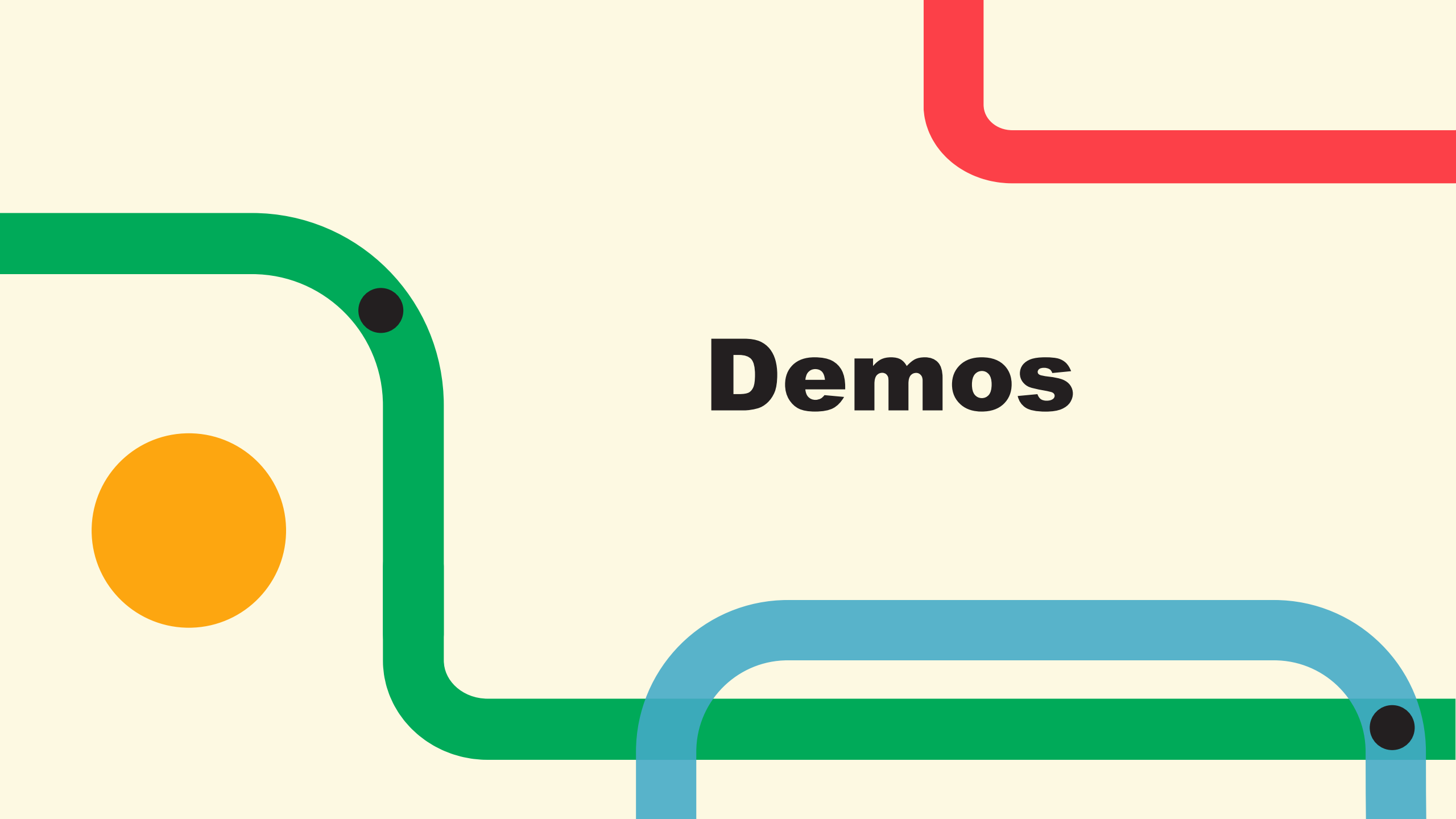


8

Sankey Imagined



College of Business: Fall 2012 first-time, first-year admitted or intended majors
(Snapshots taken on Fall Census)

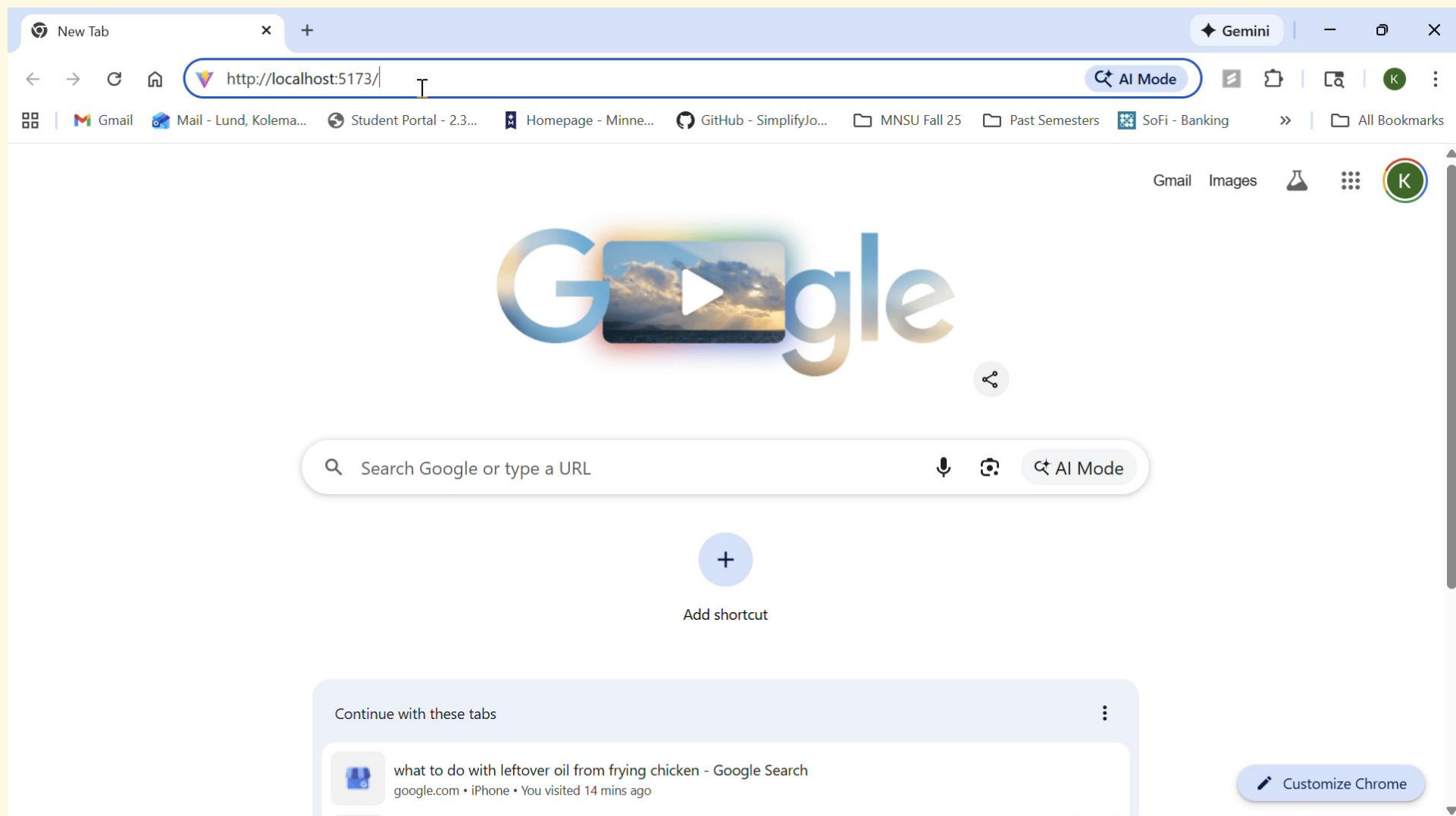
An abstract graphic design on a light cream background. It features several thick, rounded lines in green, blue, and red. A green line enters from the left, curves down, and then continues horizontally. A blue line enters from the bottom, curves up, and then continues horizontally, overlapping the green line. A red line enters from the top right and curves down. An orange circle is positioned on the left side. Two small black dots are placed on the green line: one at the first curve and one at the end of the horizontal segment.

Demos



10

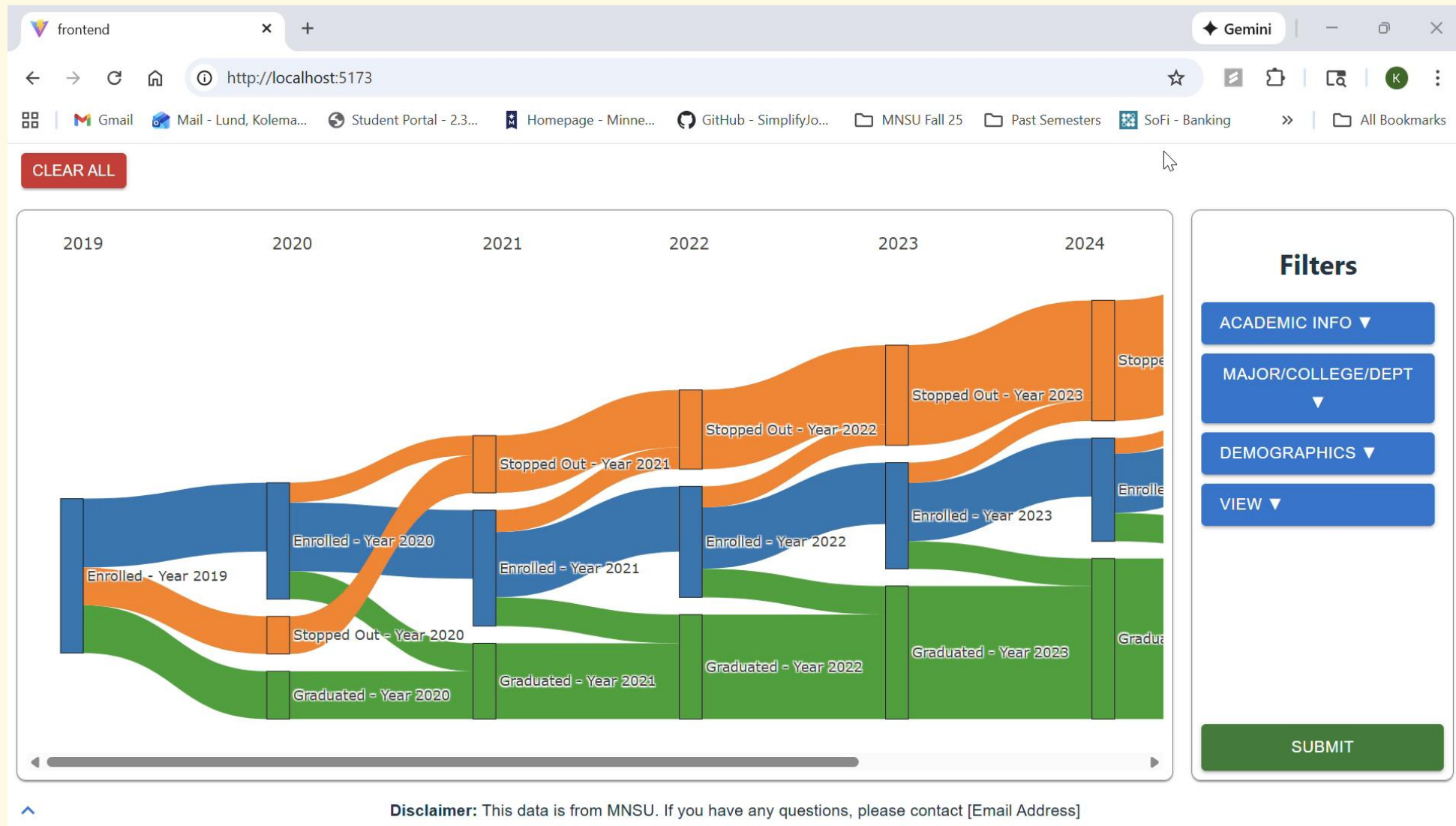
Application Home Page





11

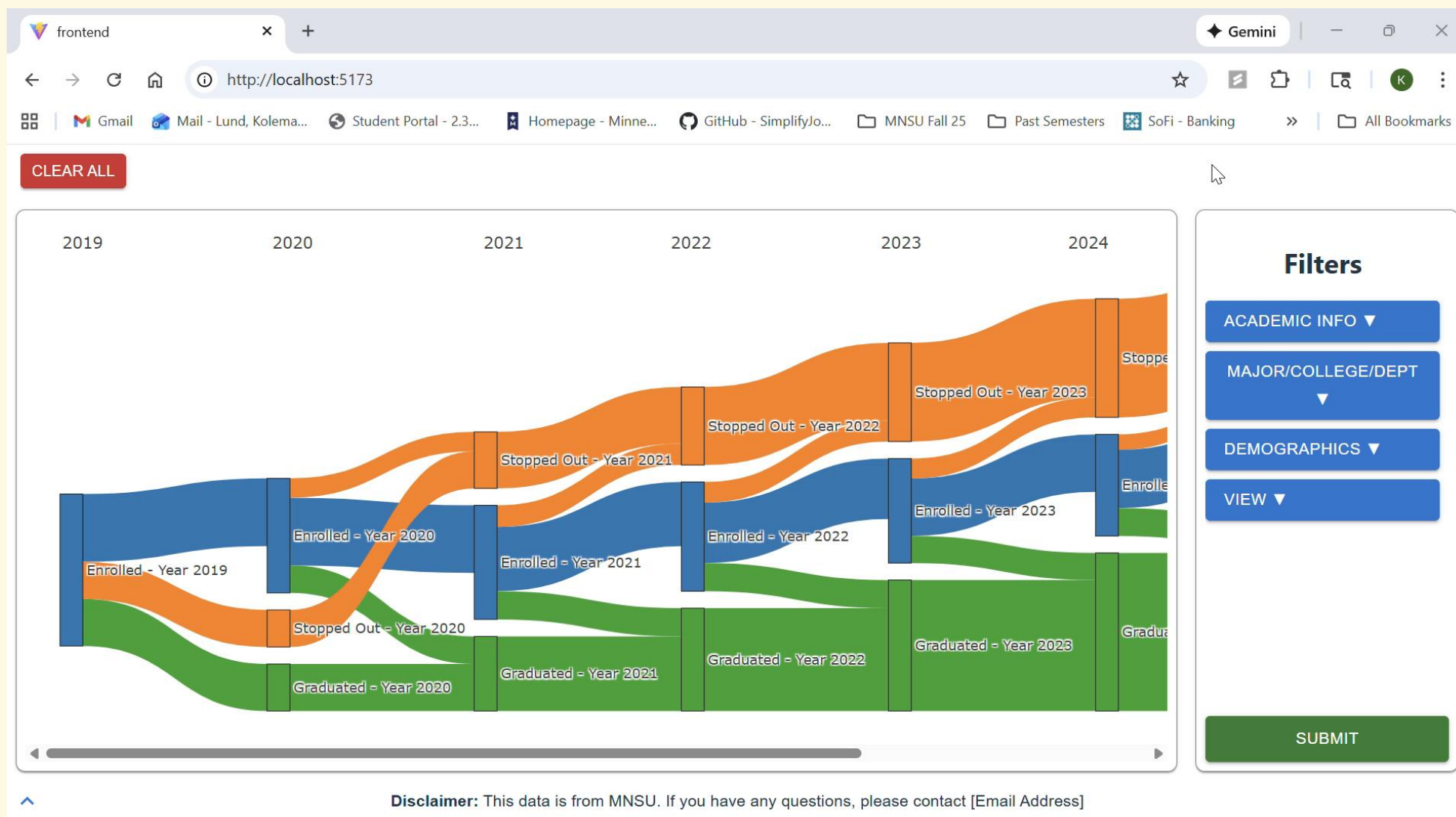
Filters Implemented





12

Different Views Created





13

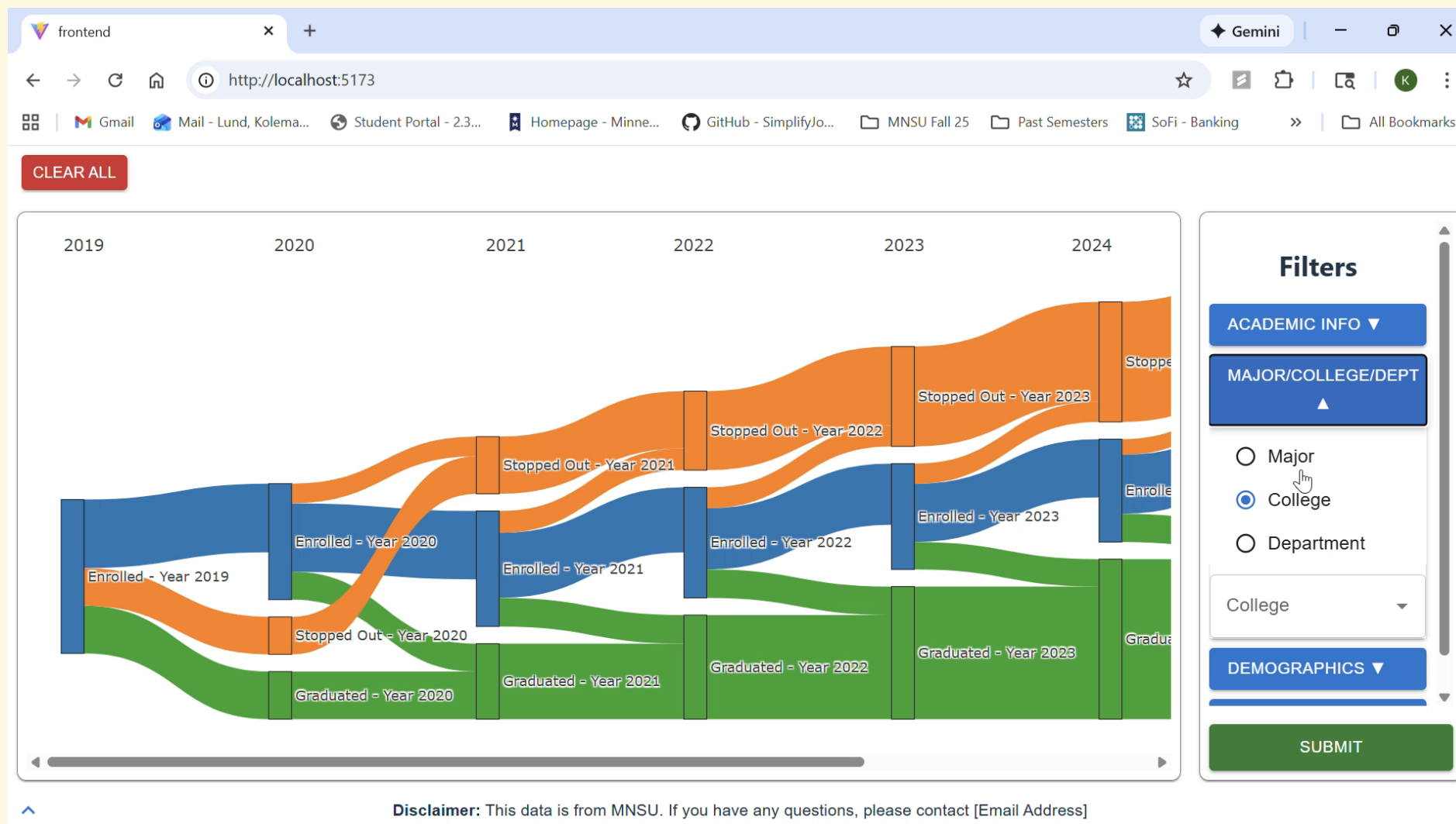
Adding Demographic Filters





14

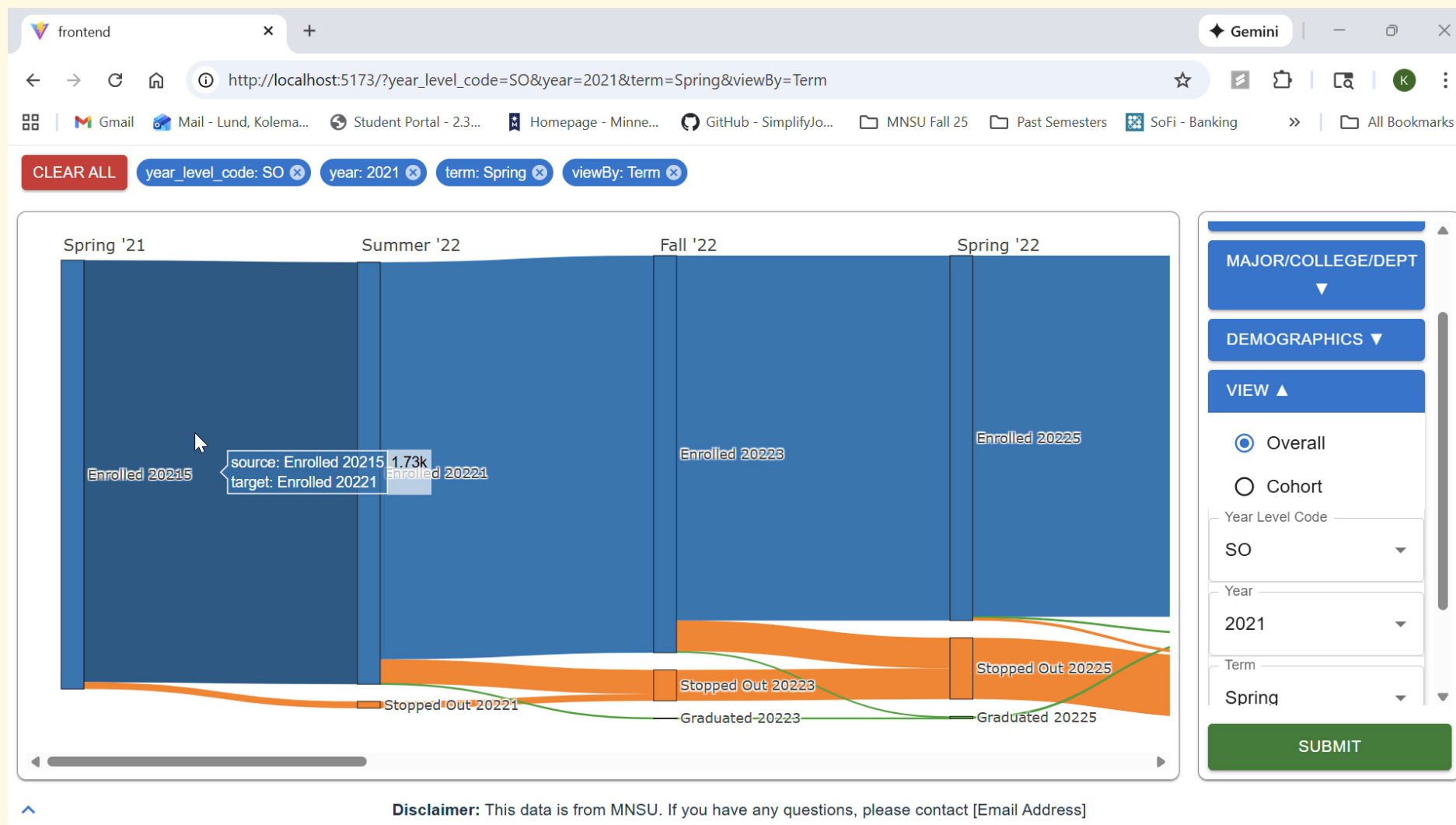
College/Major/Dept Filters





15

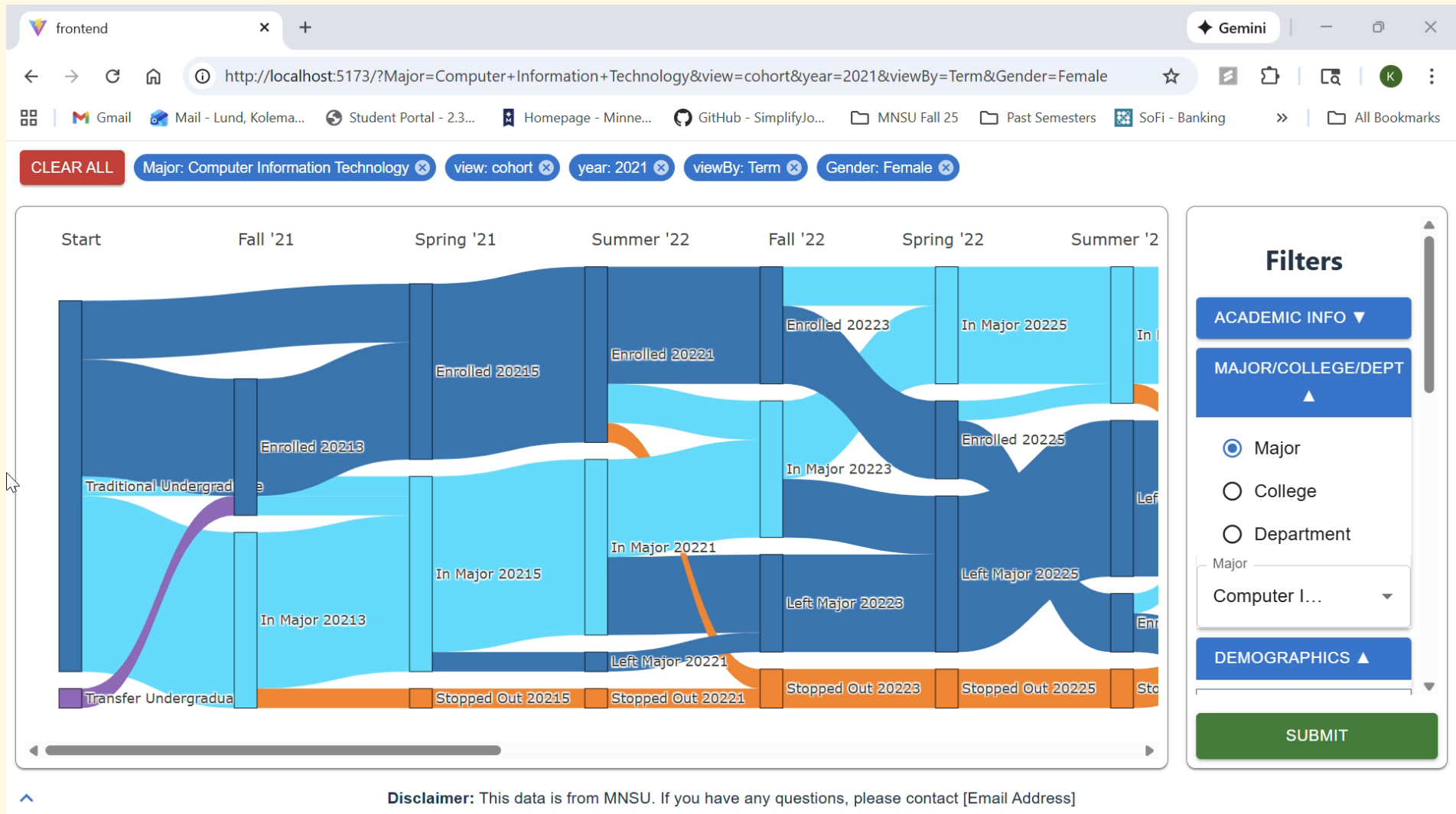
Plotly.js Features

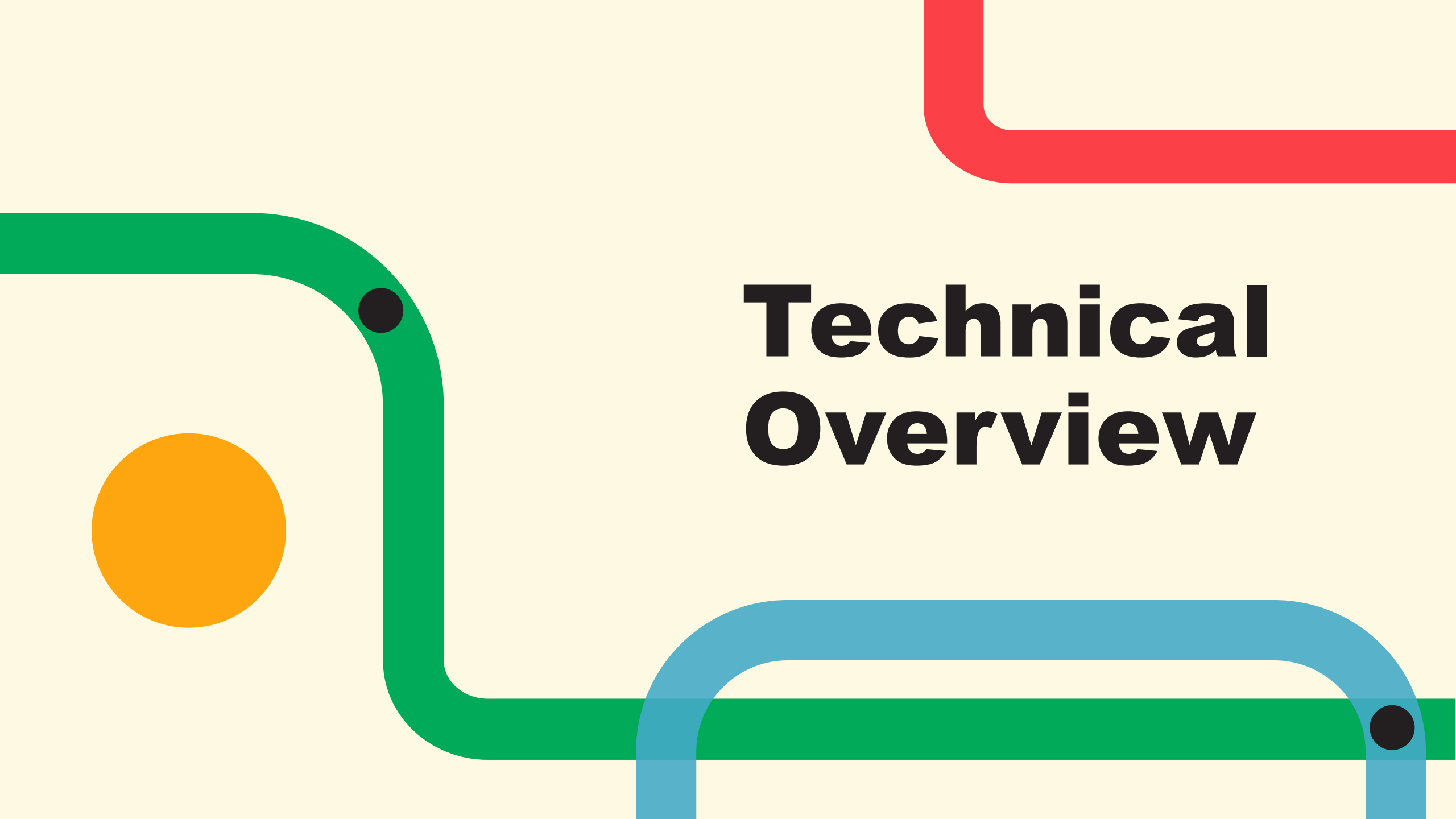




16

Sharing Views



The background features a light cream color. A thick red line enters from the top right, turns left, and extends horizontally. A thick green line enters from the left, curves down and right, then turns down and right again, crossing a thick blue line. A thick blue line enters from the bottom, curves up and left, then turns right and continues horizontally, crossing the green line. A solid orange circle is positioned on the left side. Two small black dots are located on the green line: one near the top curve and one near the bottom right end.

Technical Overview



18

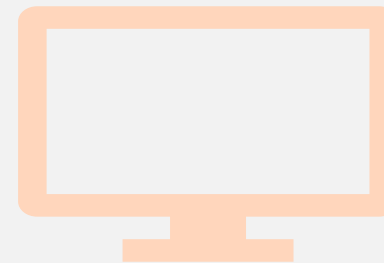
Tech Stack



**Azure
SQL
Database**



**Azure
Functions
(Python)**



**React
with
Plotly.js**



19

Database

Azure SQL



- **Azure SQL Database to store structured data**
- **Efficient retrieval and integration with Microsoft Products**



20

Data Processing

- **Received excel file with over 200,000 rows x 30 columns of student data**
- **Easy for cleaning and processing large datasets**





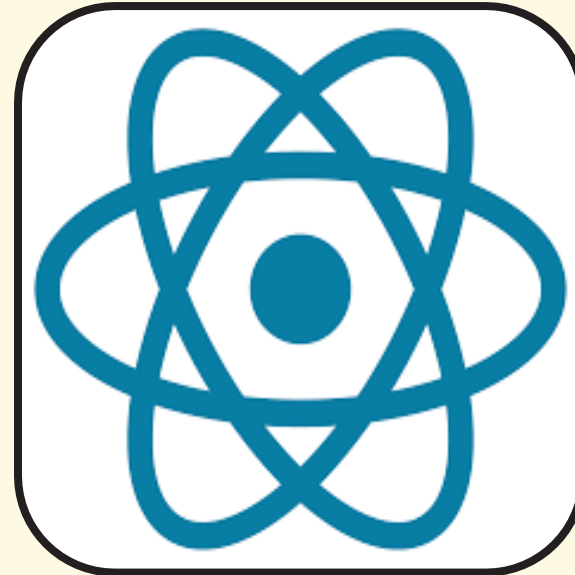
21

Serverless Python Backend to Structure Data

**Azure
Functions**



**React
Frontend**





22

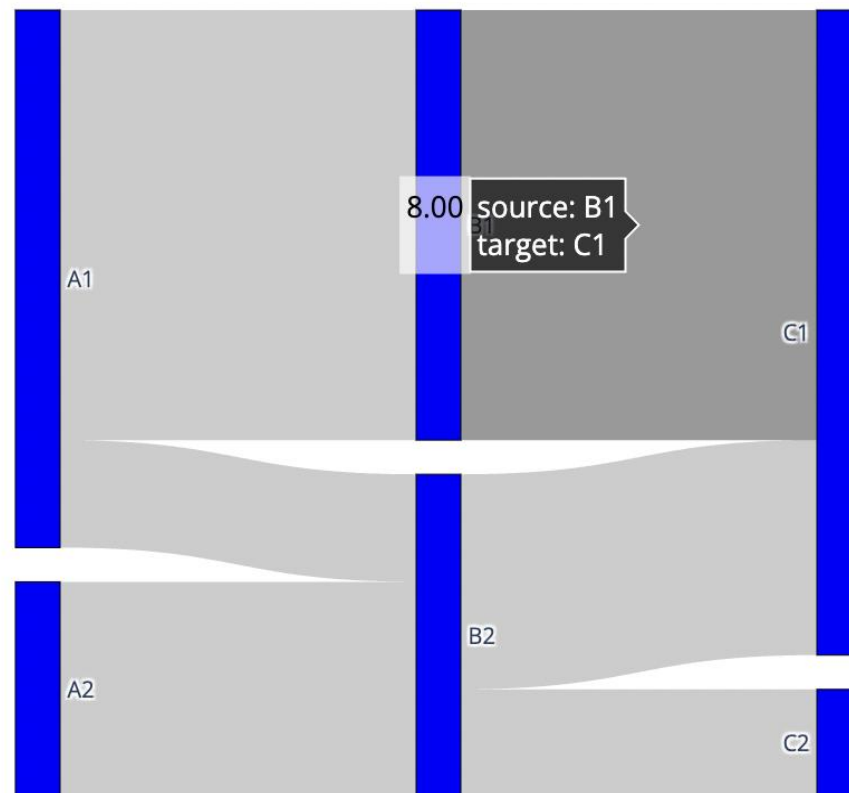
Plotly.js

Graphing library

Interactive

**Quality
Visualizations**

Basic Sankey Diagram

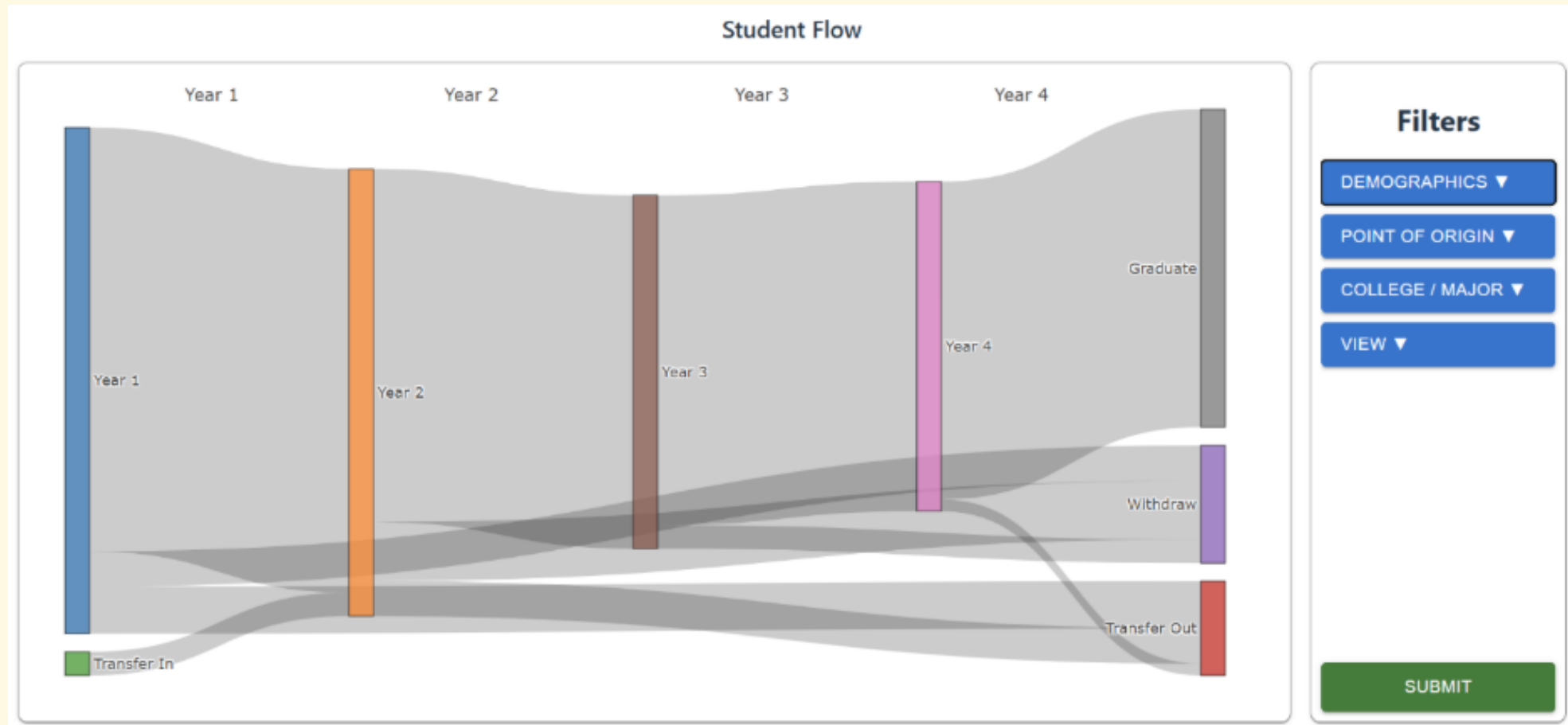


<https://plotly.com/python/sankey-diagram/>



23

React with Material UI



An abstract graphic on a light cream background. It features a thick green line that starts from the left, curves down, and then continues horizontally to the right. A thick blue line starts from the bottom, curves up, and then continues horizontally to the right, overlapping the green line. A thick red line starts from the top right and curves down to the right. A solid orange circle is positioned on the left side. Two small black dots are located on the green line: one at the first curve and one at the end of the horizontal segment on the right.

Work Completed

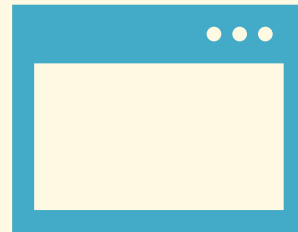


25

UI Design Document



User Stories

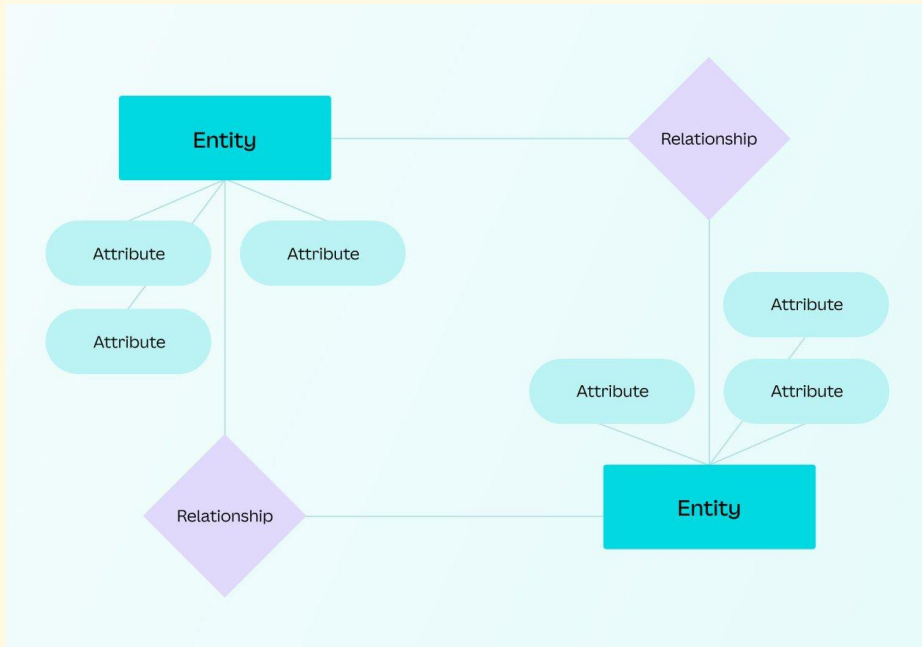


Mockups



26

Data Design Document



What data do we need?

How do we structure the data?



User Feedback Sessions

Collaborated with administrative and advising faculty to gather feedback

2.2 | Session Notes

- **Session 1 (September 25th, 2025):** Our team met with Dr. Shauna Elbers, Associate Provost for Faculty Affairs & Learning Innovation. This was prior to having any lo-fi mockups so we discussed how she would see herself using a tool like this. She mentioned two major reasons. First, it would help visualize the general patterns of students from point of contact to graduation to see possible attrition. Second, to help inform statistical models already used by faculty. It could identify variables that should be used in predictive models. She helped identify many data elements that would be useful for her and others when using this tool. A few included race, reciprocity, military service, zip code, etc. She was especially interested in including “pipeline” data, which was data about schools that we have partnerships with like South Central College or Riverland College, to view the flow of students entering MNSU from these schools. Another UI aspect she mentioned was the ability to hover over flows to see percentages of the student population. Finally, she said it would be useful to change the view of the diagram between year, semester, and even month.

2.4 | Design Adjustments Implemented

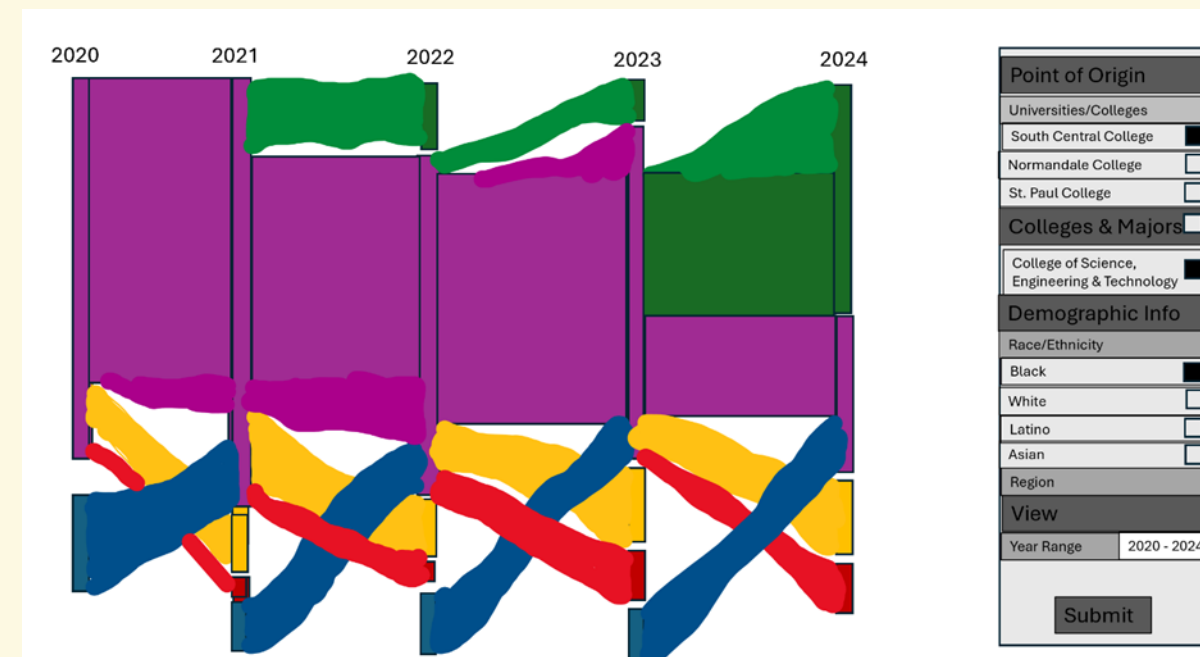
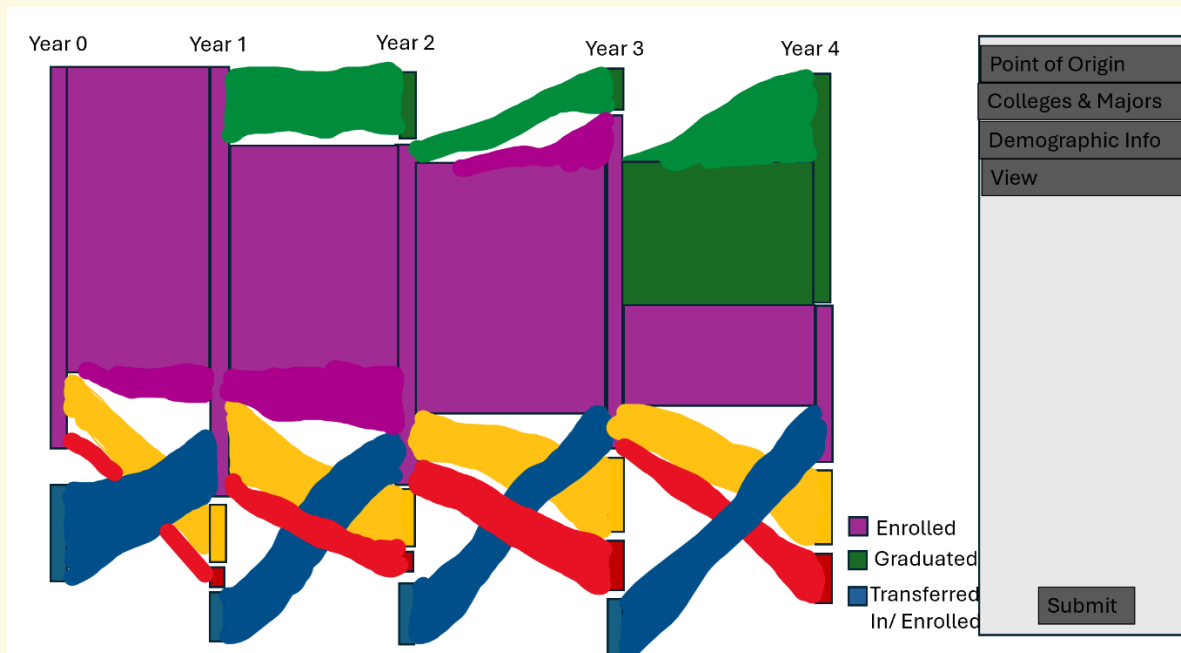
Based on the feedback received during our user feedback sessions, our team implemented or planned the following adjustments:

- Added collapsible filter categories to reduce clutter and improve navigation.
- Ensured that the package, Plotly, that we are using for the Sankey visualization has the capability to display detailed flow metrics and percentages.
- Added many of the specified data elements to our data request document.
- Revised filter list to include additional demographic and student success attributes.



28

Iterative Design

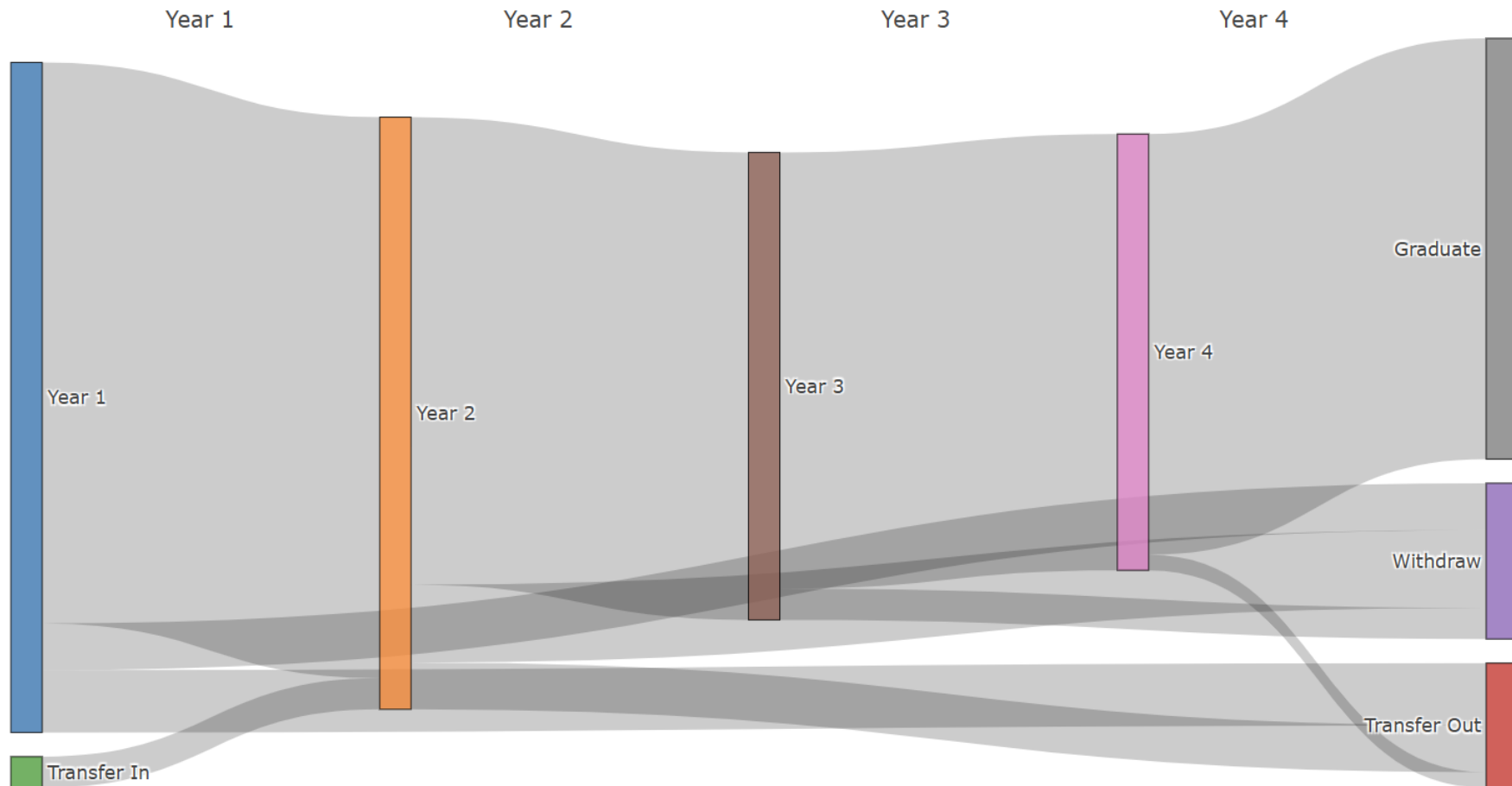




29

Prototype with Synthetic Data

Student Flow



Filters

DEMOGRAPHICS ▲

Gender ▲

✕ Clear

☐ Male

☐ Female

☐ Other

Instate Tuition ▼

Reciprocity Status ▼

Race ▼

Zip Codes ▼

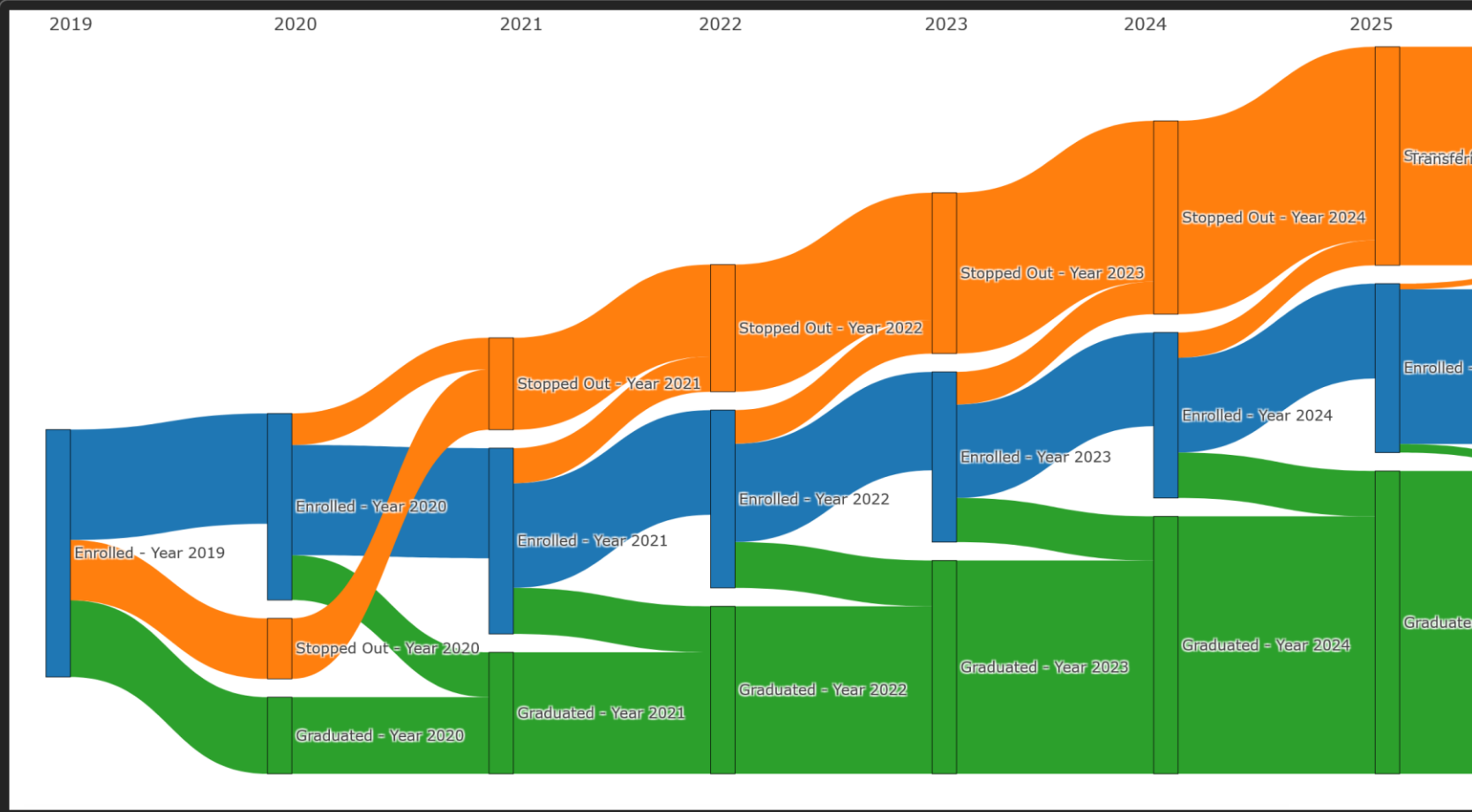
SUBMIT



30

MVP with Real Student Data

CLEAR ALL



Filters

ACADEMIC INFO ▼

MAJOR/COLLEGE/DEPT ▼

DEMOGRAPHICS ▼

VIEW ▼

SUBMIT

Disclaimer: This data is from MNSU. If you have any questions, please contact [Email Address]

The background features a light cream color with abstract geometric elements. A thick red line enters from the top right, turns 90 degrees left, and extends horizontally. A thick green line enters from the left, curves 90 degrees down, and then turns 90 degrees right to become horizontal. A thick blue line enters from the bottom, curves 90 degrees up, and then turns 90 degrees left to become horizontal. A solid orange circle is positioned on the left side. Two small black dots are located on the green line: one on the vertical segment and one on the horizontal segment. The title text is centered in the upper half of the image.

Deliverables and Timeline

**10**

Deliverables

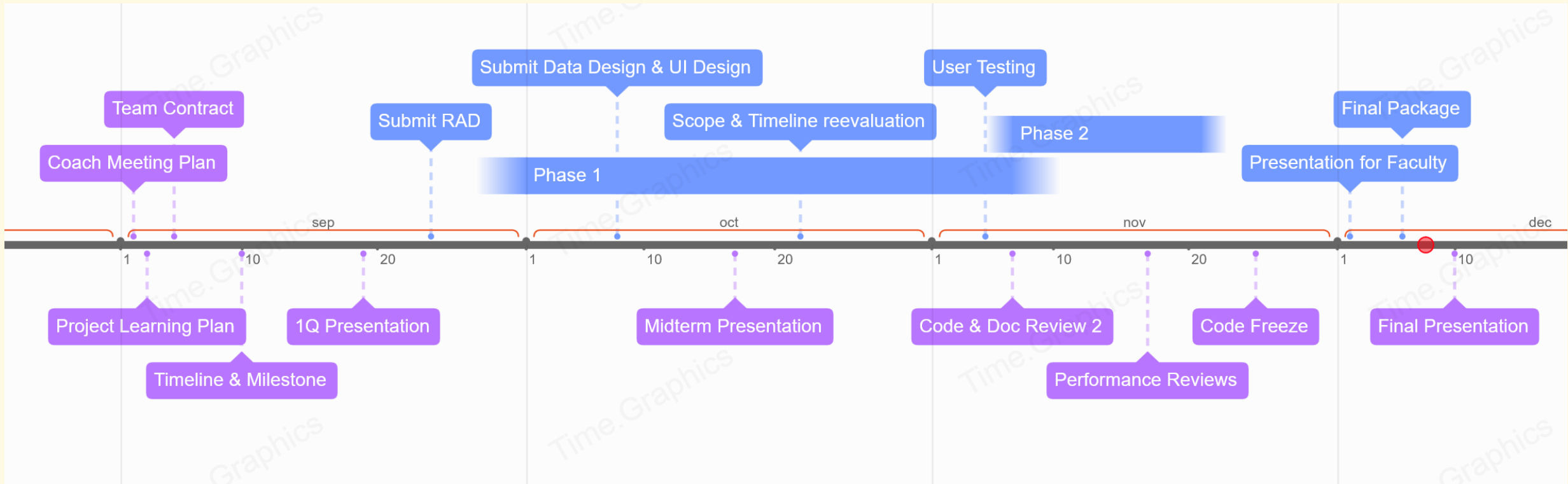
	Title of deliverable	Work completed for deliverable
	Requirements Analysis Document	Requirements gathering
	Data Design Document	Data specification, ERD diagram
	UI Design Document	UI mockups, user stories
	Software Implementation and Testing	Full stack web application, gain feedback
	Presentation to Administration Officials	Technical communication
	Handover document and digital delivery of completed work output	Documentation

	Completed on time		Late				Not completed
--	--------------------------	--	-------------	--	--	--	----------------------



33

Original Timeline



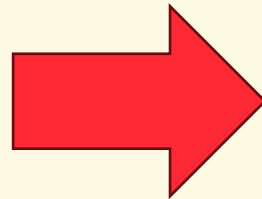


34

Two Phases

Phase 1
MNSU Data

**Rescope Date:
October 22nd**

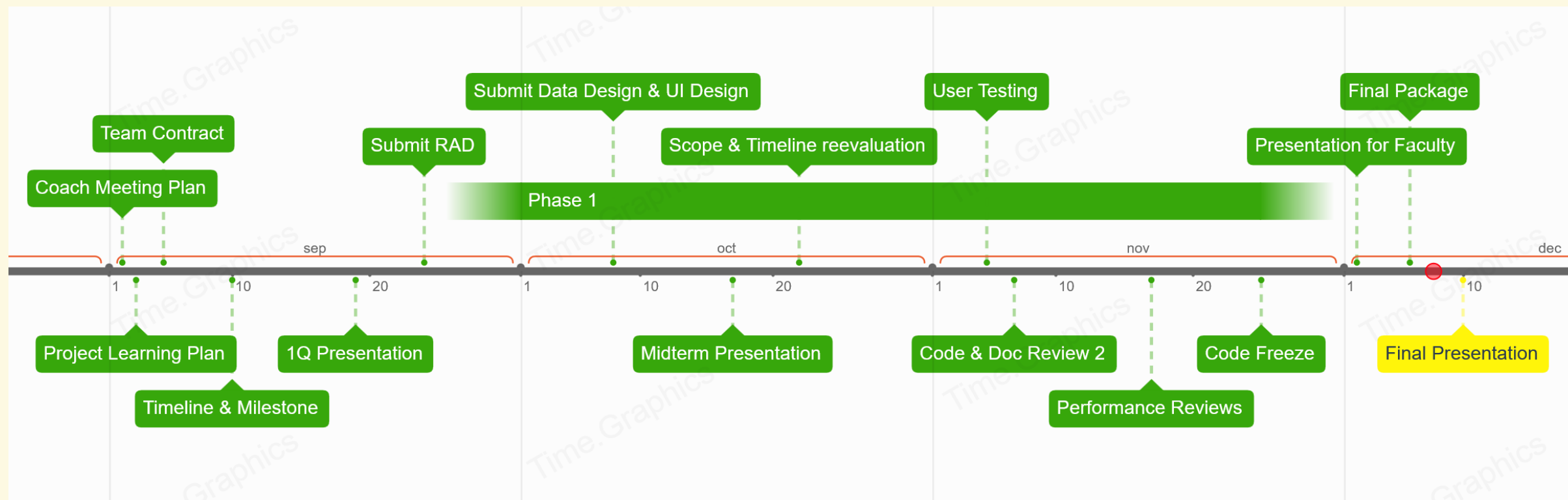


Phase 2
National Student
Clearinghouse
Data



35

Updated Timeline



Completed



In progress

The background features several abstract geometric elements: a thick red line in the top right corner, a green line that curves from the left side and then runs horizontally across the bottom, and a blue line that curves from the bottom right. There are two small black dots on the green line and one large orange circle on the left side.

Successes and Challenges



37

Successes

**Large amount
of user
feedback**

**Completing
MVP with real
student data**

**Potential
user's
excitement**



38

Communication

Client Meeting

Weekly Wed 11:00 am



zoom



Team Meeting

Tues 12:00-2:00 Weekly



Discord

Main Communications



Team Bonding

Pickleball



39

Team Bonding





40

Challenges

**Aggressive
timeline**

**Misunderstandings
on views**

Late data delivery

**41**

Client Satisfaction

Statement (Weight)	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The student team completed the project deliverables.					X
<i>1-2 sentence justification</i>	All high and medium priority items were delivered with good quality.				
The student team's overall performance was beyond expectations and could be offered as an example for future students.				X	
<i>1-2 sentence justification</i>	The team performed well both individually and as a group.				



42

Future Development

Data validation

Implement phase two data

Adding additional functionality

The background features abstract, thick, rounded lines in red and orange. On the left, a red line forms a horizontal bar with a black dot at its intersection with a vertical orange line. Below this, an orange line curves downwards. On the right, a red line curves upwards, with a black dot at its intersection with a vertical orange line. A large green circle is positioned in the lower right area, partially overlapping the red line.

Thank you

Any questions?



44

Other Graphing Libraries

- **D3.js**
 - **Not interactive**
 - **More manual coding**
- **AnyChart**
 - **Subscription fee**
- **Many others**
 - **Don't align with our client's needs**



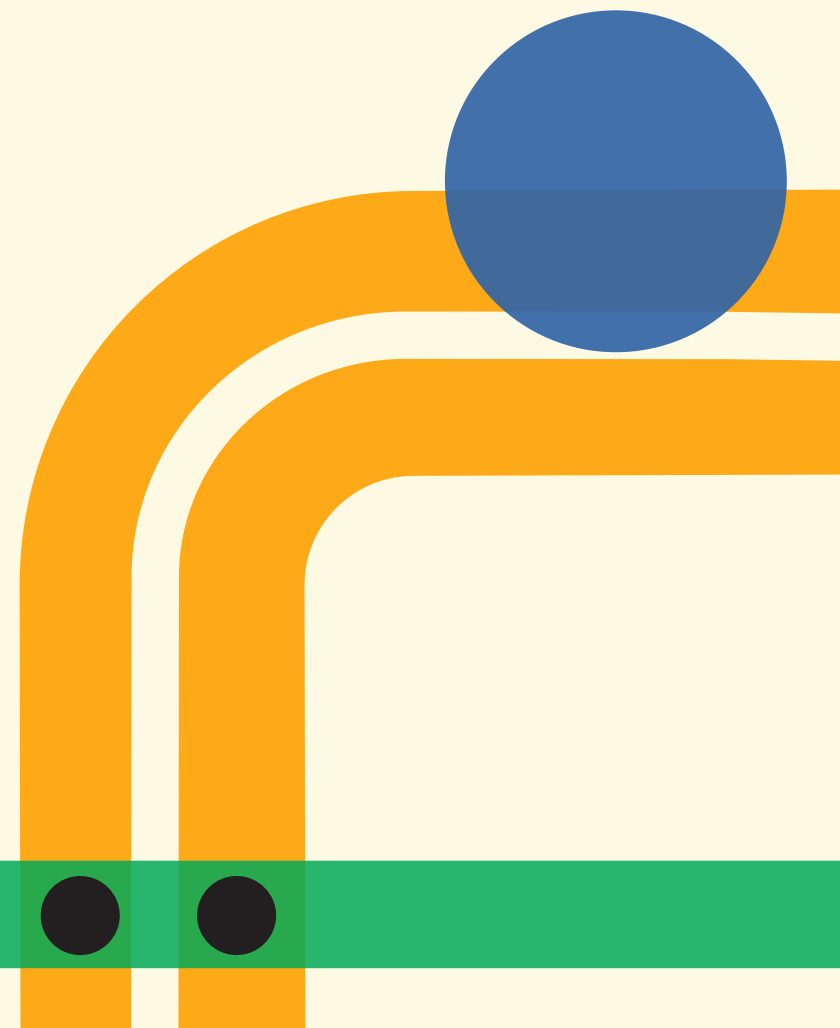
45

Deidentification of Data

Too many filters could lead to identifying individual students

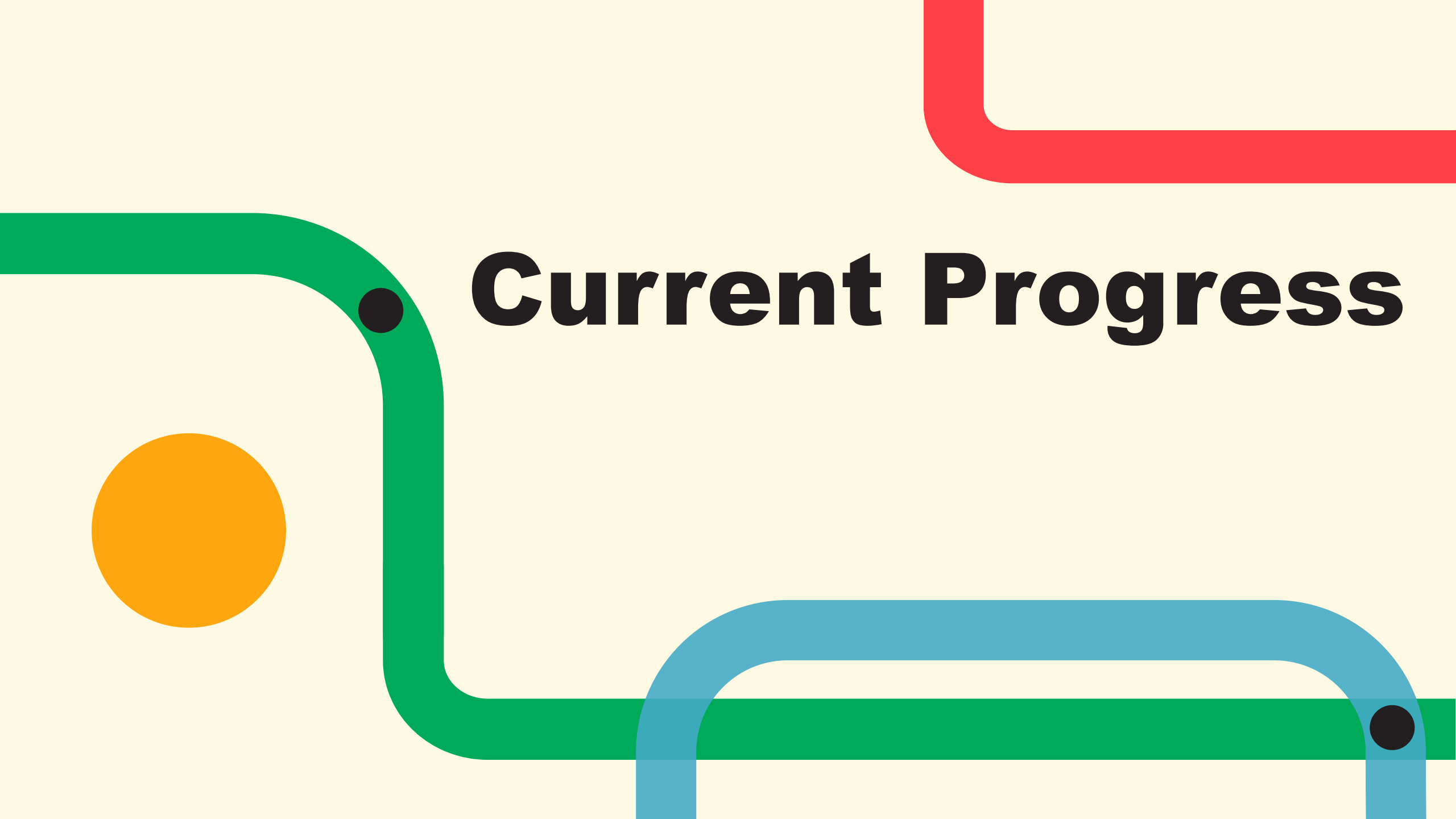
Potential filters of demographic information/school information

Could set a threshold for amount of students



An abstract graphic design on a light cream background. It features several thick, rounded lines in green, blue, and red. A green line starts from the left, curves down, and then continues horizontally. A blue line starts from the bottom, curves up, and then continues horizontally, overlapping the green line. A red line starts from the top right and curves down. There are two small black dots: one on the green line and one on the blue line. A large orange circle is positioned on the left side of the image.

Next Steps

An abstract graphic design on a light cream background. It features several thick, rounded lines in green, blue, and red. A green line starts from the left, curves down, and then continues horizontally. A blue line starts from the bottom, curves up, and then continues horizontally, overlapping the green line. A red line starts from the top right and curves down. There are two small black dots: one on the green line and one on the blue line. A large orange circle is positioned on the left side of the image.

Current Progress



49

Landing Page

Terms of Use

By continuing you confirm that you will use this application only for business purposes related to your organization. Do not use this tool to access or distribute sensitive personal data beyond what is necessary for authorized business tasks.

☐ I confirm I will use this application only for business purposes

Continue



50

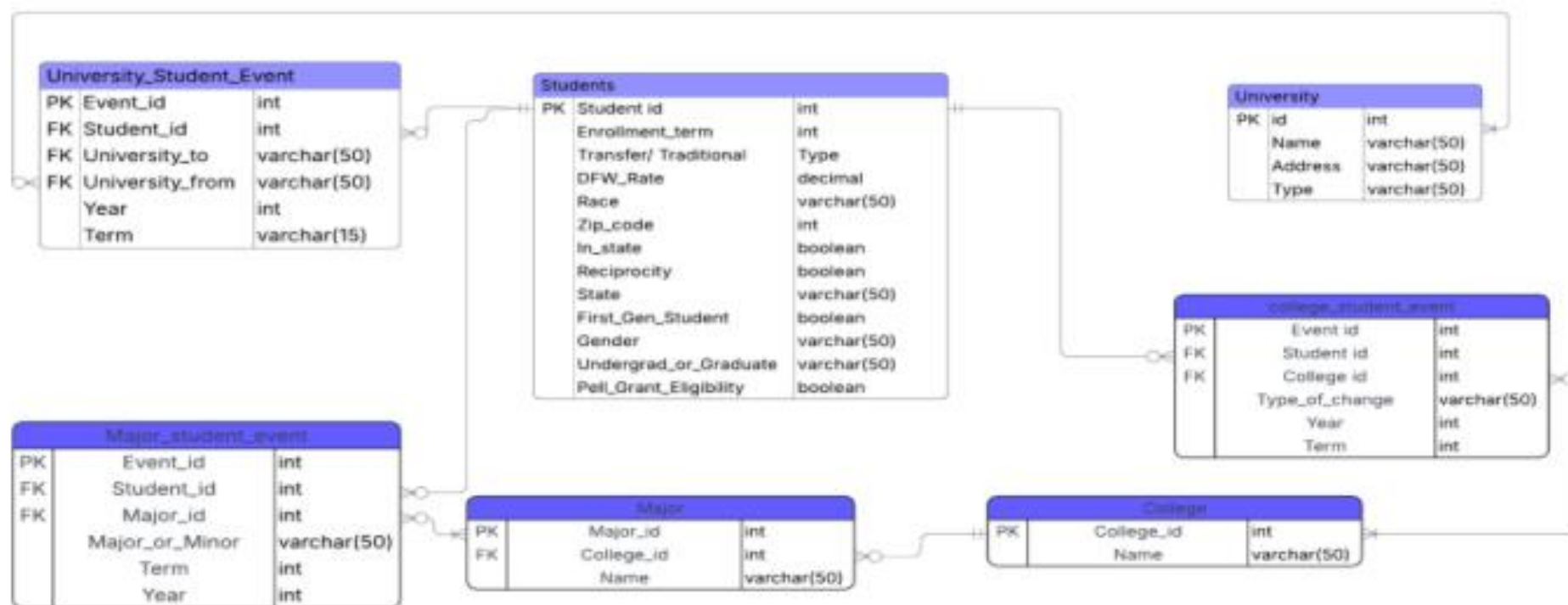
Data Request

Main Year Term Table

Field	Description	Data Type
Year Term Number	Number denoting year and term	Integer
Tech ID	Student's scrubbed tech ID	Integer
Instate	Instate Status	Boolean
Reciprocity	Reciprocity Status	Boolean
Race	Race/Ethnicity	String
Gender	Gender of student	String
State	Home state	String
Zip Code	Home zip code	Integer
Transfer or First Year Code	Code for if student transferred in or not (11 or 12)	Integer
Year Level Code	Student year level code (FR SO JR SR etc)	String
Major 1	Major 1 for student	String
Major 2	Major 2 for student	String
Major 3	Major 3 for student	String
Minor 1	Minor 1 for student	String
Minor 2	Minor 2 for student	String
Minor 3	Minor 3 for student	String
Undergrad or Graduate	Undergrad or Graduate student	String
Student Status	Part Time vs Full Time student	String
International Student	Is the student international?	Boolean
First Generation Student	Is the student a first-generation student?	Boolean
Pell Grant Eligibility	Pell Grant Eligibility Status	Boolean
DFW Rate	DFW Rate for term	Percentage (Decimal)

**51**

Schema

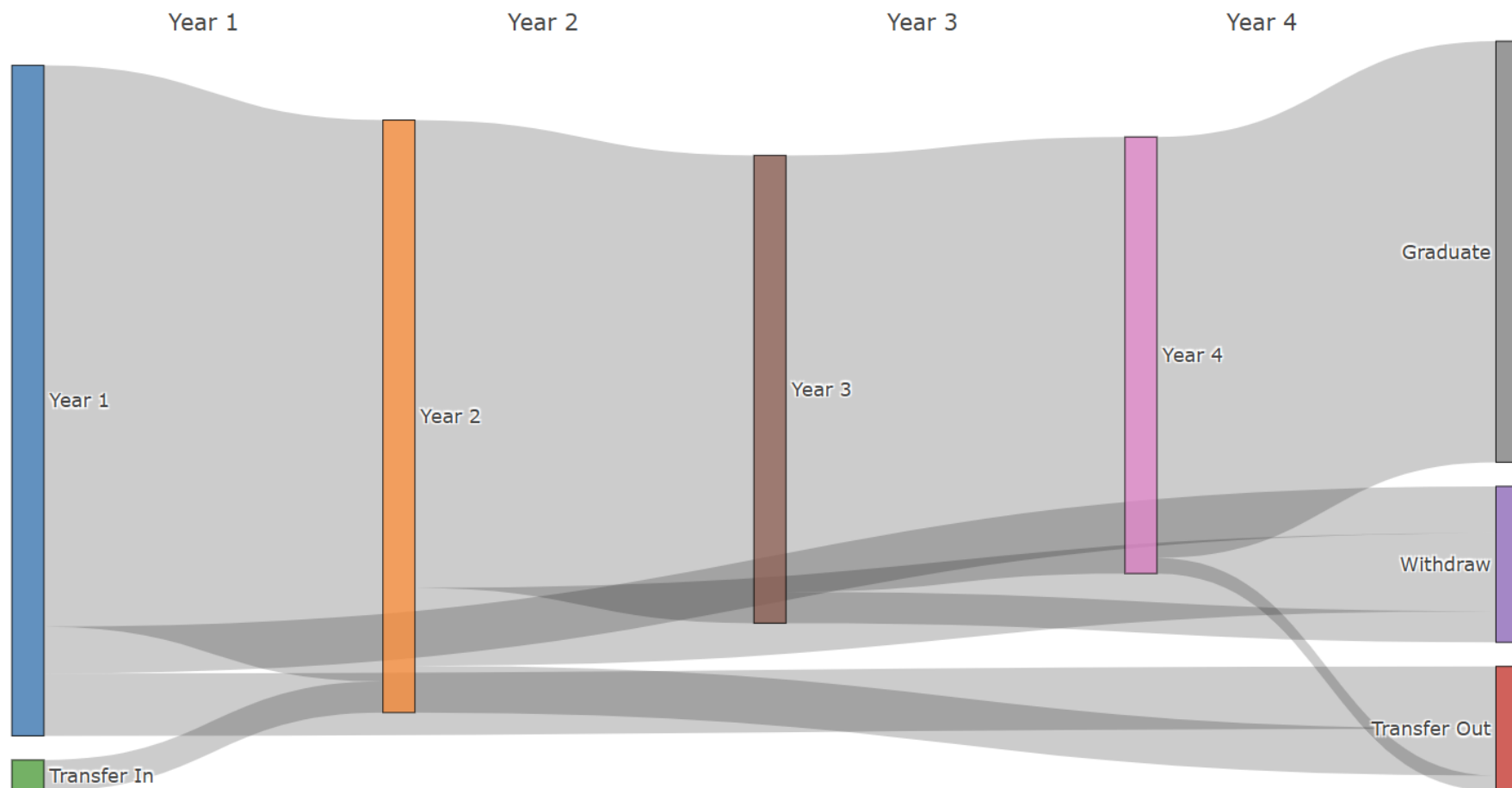




52

Prototype of Frontend Started

Student Flow



Filters

DEMOGRAPHICS ▼

POINT OF ORIGIN ▼

COLLEGE / MAJOR ▼

VIEW ▼

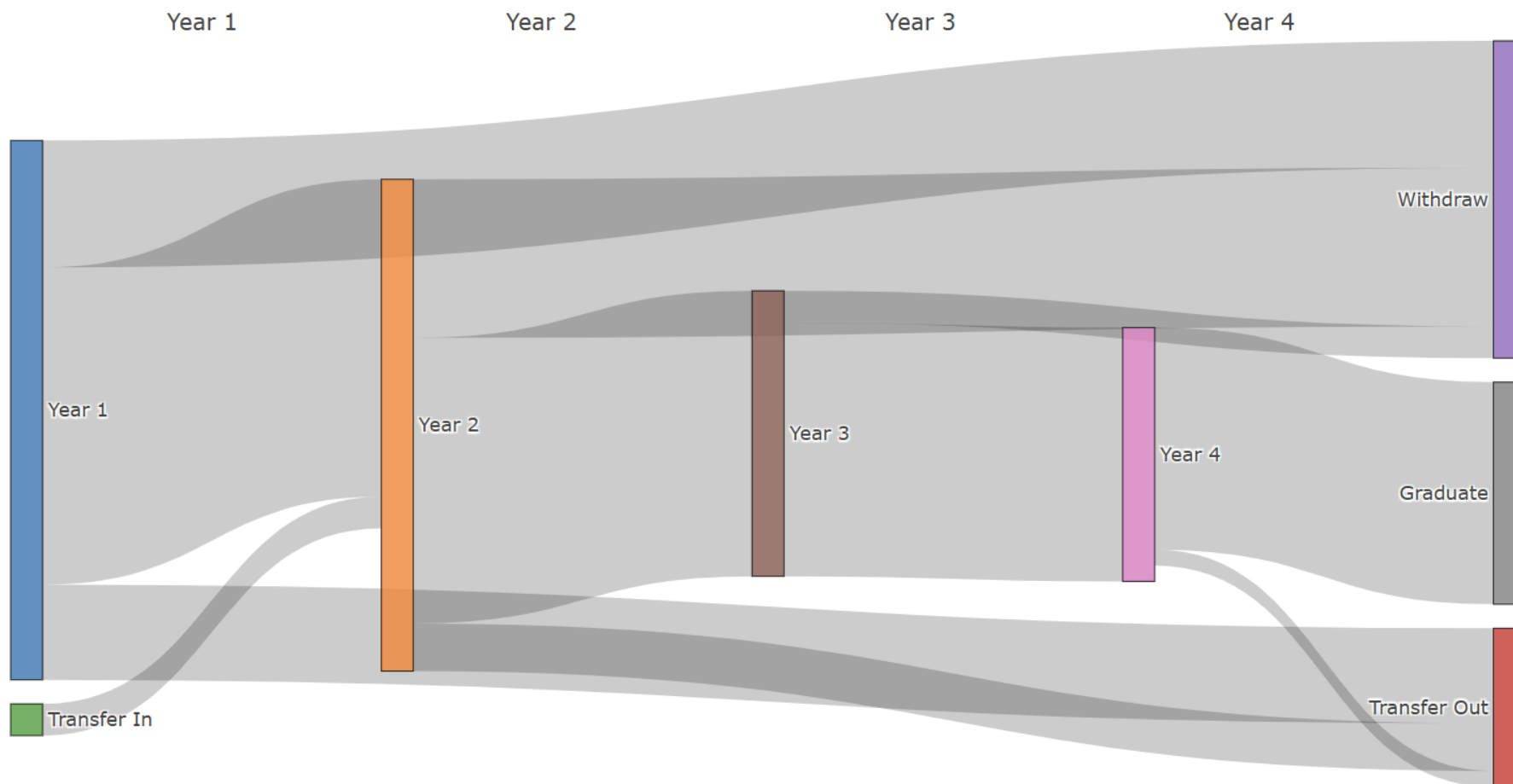
SUBMIT



53

Prototype of Frontend Started

Student Flow



Filters

DEMOGRAPHICS ▲

Gender ▲

✕ Clear

☒ Male

☐ Female

☐ Other

Instate Tuition ▲

✕ Clear

☐ True

☒ False

SUBMIT



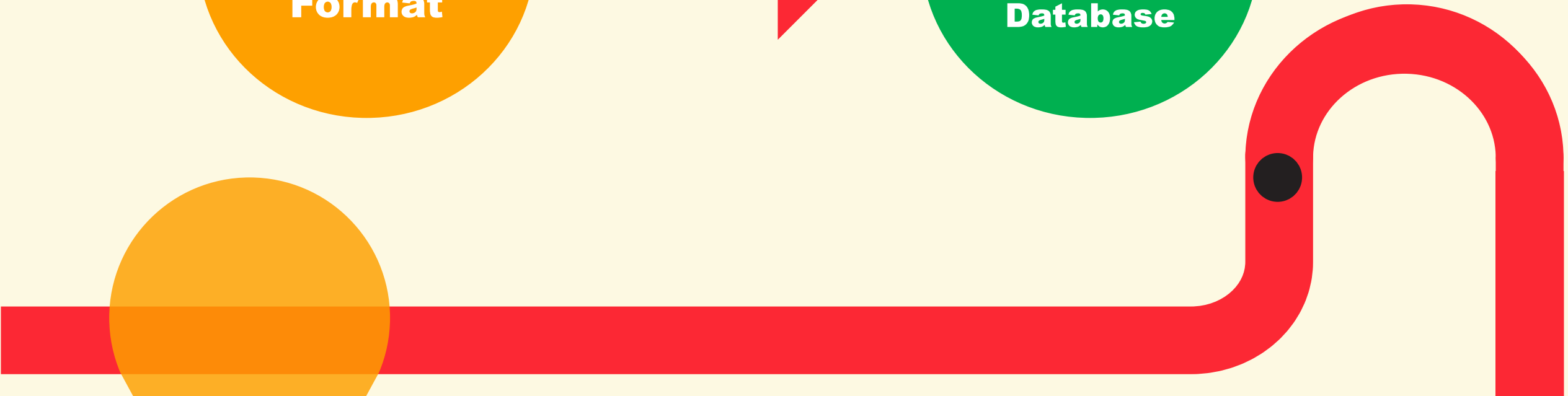
54

Data Exploration and Processing

**Raw Data
in CSV
Format**

Pandas

**Structured
Data in
Relational
Database**





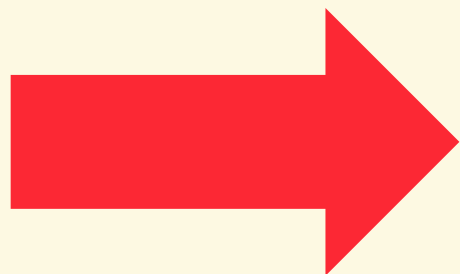
55

Add More Functional Filters

Filters

DEMOGRAPHICS ▼

SUBMIT



Filters

DEMOGRAPHICS ▼

POINT OF ORIGIN ▼

COLLEGE / MAJOR ▼

VIEW ▼

SUBMIT



56

Continue to Incorporate New Users

**Michael Bilden –
Director of
Marketing and
Recruitment**

**Abdurrahman
Guantai –
Director of
Student Access**

**Transfer
Governance
Committee
Meeting**

